

Cross-Level Residual Frequency-Guided Multi-Scale Refinement with a ConvNeXt Encoder for Polyp Segmentation

Abstract

Accurate colorectal polyp segmentation remains challenging because lesions vary in scale, morphology, contrast, and boundary appearance. Conventional multi-scale convolution blocks aggregate different receptive fields without explicitly using sample-specific frequency information from deep semantic features. We present a ConvNeXt-Tiny U-shaped framework that builds on prior Frequency-Aware Feature Enhancement (FAFEM) and Multi-Scale Convolution Block (MSCB) modules, and introduce residual frequency-guided coupling between them. The resulting RFG-MSCB reuses the low- and high-frequency descriptor produced by bottleneck FAFEM to assign sample-dependent weights to MSCB multi-kernel branches at the Stage-3 encoder-decoder fusion. The guided response is a bounded residual deviation from ordinary MSCB aggregation, preserving the conventional multi-scale path as a reference. We evaluated the model on Kvasir-SEG, CVC-ClinicDB, CVC-300, CVC-ColonDB, and ETIS-LaribPolypDB over three random seeds. The model obtained mDice/mIoU of $0.931 \pm 0.004/0.883 \pm 0.006$ on Kvasir-SEG and $0.937 \pm 0.002/0.888 \pm 0.004$ on CVC-ClinicDB. On the three external datasets, mDice ranged from 0.787 ± 0.011 on ETIS-LaribPolypDB to 0.882 ± 0.002 on CVC-300. In the raw three-seed ablation, adding residual frequency-guided coupling to FAFEM + MSCB changed Kvasir-SEG mDice from 0.930 to 0.931, an observed difference of +0.08 percentage points; mIoU changed by +0.12 percentage points, while MAE decreased slightly. The final model adds 2.4% parameters relative to FAFEM alone. The ablation therefore measures a small change under the tested protocol, while lower performance on the more challenging external datasets indicates remaining generalization limitations.

Introduction

Colorectal polyps are abnormal tissue growths that can develop into malignant lesions if they are not detected and removed at an early stage. Colonoscopy is widely used for the examination of the lower gastrointestinal tract and enables clinicians to identify and remove suspicious lesions. Nevertheless, visual inspection remains challenging because polyp appearance varies considerably across patients and imaging conditions. Variations in lesion size, morphology, texture, contrast, and location, together with specular reflections, motion blur, intestinal folds, bubbles, and other visual artifacts, can complicate accurate delineation of polyp regions. Automated polyp segmentation therefore remains an important problem in computer-aided colonoscopy, where reliable pixel-level localization may assist subsequent analysis and clinical decision support.

Accurate polyp segmentation requires a model to simultaneously capture high-level semantic information and preserve sufficiently detailed spatial representations. U-shaped encoder-decoder architectures are well suited to this task because they combine progressively abstract encoder features with higher-resolution representations through skip connections. U-Net established this general design, while subsequent architectures such as UNet++ further demonstrated the importance of multi-level feature interaction and hierarchical decoding [1, 2]. However, repeated downsampling reduces spatial resolution, and fixed decoder operations may not always provide sufficient flexibility for lesions that vary considerably in scale and visual frequency content.

Recent transformer-based and hybrid architectures have improved long-range dependency modeling in medical image segmentation. TransUNet combines convolutional representations with transformer encoding, while Polyp-PVT applies hierarchical transformer features and cross-level aggregation to polyp segmentation [3, 4]. These approaches demonstrate the importance of broad contextual modeling, but they can introduce additional architectural complexity. Modern convolutional networks such as ConvNeXt provide an alternative

by retaining the efficiency and spatial inductive bias of convolution while incorporating design principles that improve representation capacity. ConvNeXt employs large-kernel depthwise convolution, inverted channel expansion, Layer Normalization, GELU activation, layer scaling, and stochastic depth, making it a strong hierarchical encoder for dense prediction tasks [5].

Multi-scale representation is particularly important in polyp segmentation because lesions may occupy very small regions or extend over a large portion of the image. Multi-Scale Convolution Blocks (MSCBs) address this issue by combining convolutional branches with different receptive fields, allowing a network to aggregate local and broader spatial responses. However, conventional multi-branch aggregation generally combines the branch responses using a fixed formulation that is shared across all input images. Consequently, the relative contribution of different receptive fields is not explicitly conditioned on the frequency characteristics of each individual sample.

Frequency-aware feature processing provides another complementary perspective. Frequency-Aware Feature Enhancement Modules (FAFEMs) decompose deep representations into low- and high-frequency components and use these components to enhance feature responses. Low-frequency representations mainly preserve broader semantic structures, whereas high-frequency representations emphasize local spatial variations and fine structural information. Existing frequency-aware modules are typically used to improve the feature representation at the location where the frequency decomposition is performed. However, the frequency descriptor extracted from deep semantic representations can potentially provide useful global information for guiding later multi-scale processing.

This observation motivates the central idea of this work. Instead of treating bottleneck frequency enhancement and decoder multi-scale refinement as independent operations, we establish an explicit cross-level interaction between them. The frequency descriptor produced by a bottleneck FAFEM is propagated to the first decoder stage and used to regulate the relative contributions of the multi-scale branches in an MSCB. In this way, the receptive-field mixture of the decoder becomes sample dependent rather than fixed across all images.

Directly replacing the original MSCB aggregation with a fully frequency-guided formulation could, however, unnecessarily perturb a stable multi-scale representation. To address this issue, we introduce the bounded residual guidance strategy formalized in Eq. (23). Here, M_{ord} denotes the ordinary multi-scale aggregation, M_{guide} denotes the frequency-conditioned aggregation, and s is a bounded learnable guidance-strength parameter. This formulation preserves the conventional MSCB pathway as a reference and allows the model to progressively learn frequency-dependent deviations during optimization.

Based on this mechanism, we develop a ConvNeXt-Tiny U-shaped segmentation framework in which the frequency-guided modification is localized to the first encoder–decoder fusion stage; the remaining decoder stages retain conventional fusion and depthwise-separable convolution.

The principal contributions of this work are summarized as follows:

- We introduce RFG-MSCB, which couples the bottleneck FAFEM frequency descriptor to adaptive weighting of the multi-kernel MSCB branches through a bounded residual formulation. The contribution is the coupling rather than FAFEM or MSCB themselves.
- We evaluate the resulting ConvNeXt-Tiny segmentation architecture on multiple public colorectal polyp datasets and perform controlled ablation experiments to examine the contribution of frequency-aware enhancement, residual frequency guidance, and the selected Stage-3 placement.
- We quantify the architectural trade-off: the localized RFG-MSCB modification increases the parameter count by 2.4% relative to the FAFEM-only configuration under the reported complexity convention.

Related work

Encoder–decoder architectures for medical image segmentation

Fully convolutional encoder–decoder networks remain a central design pattern for biomedical image segmentation. U-Net introduced a symmetric contracting and expanding architecture in which high-resolution encoder

features are transferred to the decoder through skip connections, allowing semantic context and spatial detail to be combined during prediction [1]. UNet++ further showed that the structure of skip pathways is important by using nested dense connections to reduce the semantic discrepancy between encoder and decoder representations [2]. These works established the importance of hierarchical feature reuse, but they do not prescribe how the decoder should adapt its receptive-field mixture to the image-specific content of a polyp.

Polyp-specific convolutional models have explored efficient alternatives to large attention-based designs. HarDNet-MSEG combines a low-memory-traffic convolutional backbone with a lightweight decoder and reports strong polyp segmentation can be obtained without a transformer-heavy architecture [6]. PraNet introduced a parallel partial decoder and reverse attention to progressively refine uncertain polyp regions [7]. These methods highlight the value of efficient decoding and iterative refinement, but their main mechanisms are not designed to use frequency information from the bottleneck to regulate multi-scale convolutional branches at an earlier decoder stage.

Modern convolutional encoders provide another route to high-capacity dense prediction. ConvNeXt revisits convolutional network design with large-kernel depthwise convolution, inverted channel expansion, Layer Normalization, GELU activation, layer scaling, and stochastic depth, producing a hierarchical backbone with strong representation capacity [5]. We use ConvNeXt-Tiny as the encoder and focus the new mechanism on the interaction between bottleneck frequency information and Stage-3 multi-scale convolution.

Transformer and contextual polyp segmentation

The large variation in polyp size, shape, color, and boundary contrast motivates architectures that capture both local appearance and broader anatomical context. Transformer-based approaches address this need by modeling long-range spatial relationships. TransUNet combines convolutional feature extraction with a transformer encoder for medical image segmentation [3], and Polyp-PVT applies pyramid vision transformer features and specialized fusion to polyp segmentation [4]. Recent polyp-segmentation studies have continued this direction through contrastive transformer modeling, contextual information flow, phase-wise feature pyramids, and pyramid-transformer feature fusion [8, 9, 10, 11].

These approaches show the value of global context and cross-level interaction, while motivating a convolutional alternative to more complex attention-based designs.

Efficient multi-scale convolutional decoding

Multi-scale convolution is widely used because polyps may appear as small, subtle lesions or occupy a large image region. Parallel kernels and dilated or depthwise branches allow a decoder to aggregate local and broader responses without relying entirely on self-attention. EMCAD introduced an efficient multi-scale convolutional attention decoder for medical image segmentation and uses Multi-Scale Convolution Blocks (MSCBs) as part of its lightweight decoding design [12]. In an MSCB, multiple convolutional branches provide different receptive fields and are combined to form a multi-scale response.

MSCB provides the multi-scale reference path used by the present comparison.

Frequency-aware and adaptive feature recalibration

Attention and recalibration mechanisms provide a general precedent for using compact descriptors to control feature responses. Squeeze-and-Excitation Networks use global channel descriptors to recalibrate convolutional features [13], while Selective Kernel Networks adaptively weight convolutional branches with different receptive fields [14]. These methods are important novelty comparators because they show that descriptor-driven reweighting and adaptive receptive-field selection are established ideas.

Frequency-aware attention provides a closer conceptual link. FcaNet interprets channel attention through frequency analysis and uses frequency components for channel attention [15]. In polyp segmentation, Polyp-Mamba combines multi-frequency perception with a gated selection mechanism to model frequency-sensitive

representations [16]. These studies motivate frequency-aware recalibration, while the present work applies that idea to the relationship between bottleneck enhancement and decoder refinement.

Boundary and recent polyp-segmentation refinements

Weak and visually ambiguous lesion boundaries remain a major challenge in polyp segmentation. Boundary-aware methods often introduce explicit edge supervision, auxiliary contour branches, image-pyramid high-frequency cues, or reverse attention. MEGANet employs multi-scale edge-guided attention for weak-boundary polyp segmentation [17], and MSBP-Net performs multi-scale boundary prediction to improve contour delineation [18]. Other recent methods model uncertainty, contextual flow, frequency-sensitive representations, and multi-stage refinement [19, 9, 16, 20].

The network does not add an explicit boundary branch or edge label; it uses the boundary-aware loss described in the Experiments section. Boundary-specific improvement is not evaluated separately, so the frequency-guided decoder should not be described as a dedicated boundary module.

Positioning of the method

Existing work has improved polyp segmentation through U-shaped skip reuse, reverse attention, transformer-based context modeling, efficient convolutional decoding, frequency-aware representation learning, and boundary-aware refinement. The method is positioned at the intersection of efficient multi-scale convolution and frequency-aware recalibration.

Specifically, the final model combines a ConvNeXt-Tiny encoder, bottleneck frequency enhancement, and efficient multi-scale decoding. We adopt the FAFEM and MSCB structures; the new contribution is their cross-level coupling at the Stage-3 fusion, which is developed technically in the next section.

Method

Overall Architecture

The network uses a U-shaped encoder–decoder architecture built upon a ConvNeXt-Tiny encoder. Frequency information extracted from deep semantic features regulates multi-scale processing during decoding. Frequency enhancement and multi-scale convolution are linked through a cross-level connection between the bottleneck and the first decoder stage.

Let $I \in \mathbb{R}^{3 \times H \times W}$ denote an input colonoscopy image. The ConvNeXt-Tiny encoder extracts four hierarchical feature maps,

$$\{F_1, F_2, F_3, F_4\} = E(I), \quad (1)$$

where $F_1 \in \mathbb{R}^{96 \times H/4 \times W/4}$, $F_2 \in \mathbb{R}^{192 \times H/8 \times W/8}$, $F_3 \in \mathbb{R}^{384 \times H/16 \times W/16}$, and $F_4 \in \mathbb{R}^{768 \times H/32 \times W/32}$. For an input resolution of 352×352 , their spatial dimensions are 88×88 , 44×44 , 22×22 , and 11×11 , respectively.

The deepest encoder feature F_4 is first processed by a Frequency-Aware Feature Enhancement Module (FAFEM), which produces an enhanced bottleneck representation together with a compact frequency descriptor. The enhanced representation is passed through a lightweight residual bottleneck, whereas the frequency descriptor is propagated to the first decoder stage. After the decoder feature is fused with F_3 , a Residual Frequency-Guided Multi-Scale Convolution Block (RFG-MSCB) uses the frequency descriptor to adaptively regulate its multi-scale convolution branches.

The overall processing can therefore be summarized as

$$(\tilde{F}_4, \mathbf{z}_f) = \mathcal{F}_{\text{FAFEM}}(F_4), \quad (2)$$

$$B = \mathcal{B}(\tilde{F}_4), \quad (3)$$

$$D_4 = \mathcal{M}_{\text{RFG}}(\mathcal{S}_3[\mathcal{D}_4(B), F_3], \mathbf{z}_f). \quad (4)$$

followed by

$$D_3 = \mathcal{S}_2[\mathcal{D}_3(D_4), F_2], \quad (5)$$

$$D_2 = \mathcal{S}_1[\mathcal{D}_2(D_3), F_1], \quad (6)$$

$$D_1 = \mathcal{D}_1(D_2). \quad (7)$$

Here, $\mathcal{D}_i$ denotes a decoder block, $\mathcal{S}_i$ denotes the corresponding skip-feature fusion operation, $\mathbf{z}_f$ is the frequency descriptor extracted at the bottleneck, and $\mathcal{M}_{\text{RFG}}$ denotes the residual frequency-guided multi-scale refinement mechanism. Equations (4)–(7) summarize the decoder path.

The final prediction head transforms D_1 into a single-channel segmentation logit map, which is bilinearly resized to the original image resolution.

ConvNeXt-Tiny Encoder

The encoder employs ConvNeXt-Tiny with stage depths $[3, 3, 9, 3]$ and channel dimensions $[96, 192, 384, 768]$. Before entering the encoder, each RGB image is normalized using the ImageNet statistics $\boldsymbol{\mu} = (0.485, 0.456, 0.406)$ and $\boldsymbol{\sigma} = (0.229, 0.224, 0.225)$.

The first stage uses a 4×4 convolution with stride 4 to generate F_1 . Each subsequent stage begins with spatial downsampling using a 2×2 stride-2 convolution.

For an input feature X , a ConvNeXt block first performs a 7×7 depthwise convolution. The resulting feature is normalized in channels-last format and processed by a pointwise multilayer perceptron with a fourfold channel expansion:

$$\begin{aligned} Z &= \text{DWConv}_{7 \times 7}(X), \\ V &= W_2(\text{GELU}(W_1(\text{LN}(Z)))) \end{aligned} \quad (8)$$

The block output is obtained through layer-scaled residual learning,

$$Y = X + \text{DropPath}(\gamma \odot V), \quad (9)$$

where γ is a learnable layer-scale parameter initialized to 10^{-6} .

A stochastic-depth rate of 0.25 is used throughout the encoder. Feature dropout with probability 0.2 is additionally applied after the second, third, and fourth encoder stages.

Frequency-Aware Bottleneck Enhancement

The deepest encoder representation F_4 contains strong semantic information but has undergone substantial spatial downsampling. To expose complementary frequency characteristics of this representation, FAFEM is applied exclusively at the bottleneck.

Given $X = F_4$, a local low-frequency approximation is obtained using 3×3 average pooling,

$$X_L = \text{AvgPool}_{3 \times 3}(X), \quad (10)$$

$$X_H = X - X_L. \quad (11)$$

The low- and high-frequency components in Eqs. (10) and (11) are independently transformed using depthwise convolutional branches,

$$\hat{X}_L = \phi_L(X_L), \quad \hat{X}_H = \phi_H(X_H), \quad (12)$$

where each $\phi(\cdot)$ contains a 3×3 depthwise convolution followed by GELU activation.

The two components are concatenated and globally pooled to obtain the frequency descriptor

$$\mathbf{z}_f = \text{GAP}([\hat{X}_L, \hat{X}_H]) \in \mathbb{R}^{2C}. \quad (13)$$

Since $C = 768$ at the bottleneck, $\mathbf{z}_f \in \mathbb{R}^{1536}$.

A lightweight channel-gating network generates adaptive weights for the low- and high-frequency representations. Their weighted combination is spatially refined and introduced through a residual connection:

$$\tilde{F}_4 = F_4 + \gamma_f \mathcal{R}_f(w_L \odot \hat{X}_L + w_H \odot \hat{X}_H), \quad (14)$$

where γ_f is a learnable scalar initialized to zero.

Lightweight Residual Bottleneck

The frequency-enhanced representation $\tilde{F}_4$ is subsequently processed by a lightweight residual bottleneck.

A 1×1 convolution first reduces the number of channels from 768 to 192:

$$X_r = \text{GELU}\left(\text{LN}\left(\text{Conv}_{1 \times 1}(\tilde{F}_4)\right)\right). \quad (15)$$

A 3×3 depthwise-separable convolution performs spatial transformation in the reduced feature space. The representation is subsequently projected back to 768 channels.

The bottleneck output is formulated as

$$B = \tilde{F}_4 + \gamma_b \mathcal{H}(\tilde{F}_4), \quad (16)$$

where $\mathcal{H}(\cdot)$ denotes the bottleneck transformation and γ_b is a learnable scalar initialized to zero.

This initialization makes the bottleneck behave initially as an identity mapping and allows its transformation to be progressively introduced during optimization.

Residual Frequency-Guided Multi-Scale Convolution

The principal architectural contribution of this work is the Residual Frequency-Guided Multi-Scale Convolution mechanism.

After the bottleneck feature is upsampled from $H/32$ to $H/16$, the decoder representation is reduced from 768 to 384 channels and concatenated with the Stage-3 encoder feature F_3 . A lightweight fusion block transforms the resulting 768-channel representation into $X_3 \in \mathbb{R}^{384 \times H/16 \times W/16}$, or $\mathbb{R}^{384 \times 22 \times 22}$ for a 352×352 input.

A conventional lightweight multi-scale convolution block first expands the feature dimension from C to $2C$:

$$X_e = \text{ReLU6}(\text{BN}(\text{Conv}_{1 \times 1}(X_3))). \quad (17)$$

Three parallel depthwise convolution branches with kernel sizes $k \in \{1, 3, 5\}$ then generate multi-scale responses,

$$B_k = \phi_k(X_e), \quad (18)$$

where each ϕ_k consists of a depthwise convolution, Batch Normalization, and ReLU6 activation.

In a conventional MSCB, these branches are combined uniformly:

$$M_{\text{ord}} = B_1 + B_3 + B_5. \quad (19)$$

The new mechanism conditions the relative contributions of the three receptive fields on the bottleneck frequency descriptor $\mathbf{z}_f$.

A lightweight guidance network maps the 1536-dimensional descriptor to three branch logits:

$$\mathbf{q} = W_{g2}(\text{GELU}(W_{g1}\mathbf{z}_f)), \quad (20)$$

where $W_{g1} : 1536 \rightarrow 48$ and $W_{g2} : 48 \rightarrow 3$.

The normalized branch weights are calculated as

$$[\alpha_1, \alpha_3, \alpha_5] = 3 \text{Softmax}(\mathbf{q}), \quad (21)$$

such that $\alpha_1 + \alpha_3 + \alpha_5 = 3$.

The frequency-guided response becomes

$$M_{\text{guide}} = \alpha_1 B_1 + \alpha_3 B_3 + \alpha_5 B_5. \quad (22)$$

We introduce the guidance as a bounded residual deviation from the conventional multi-scale response:

$$M_{\text{RFG}} = M_{\text{ord}} + s(M_{\text{guide}} - M_{\text{ord}}), \quad (23)$$

where $0 \leq s \leq 0.5$.

The learnable guidance-strength parameter s is initialized to 0.05.

The final guidance-layer weights and biases are initialized so that the branch logits are equal. Thus, at initialization, $\alpha_1 = \alpha_3 = \alpha_5 = 1$ and $M_{\text{guide}} = M_{\text{ord}}$.

The module thus initially reproduces the conventional multi-scale path, while optimization is allowed to learn image-dependent deviations from uniform branch weighting.

The resulting feature is processed using channel shuffle and a 1×1 projection from $2C$ back to C , followed by Batch Normalization. A final residual connection preserves the input representation:

$$Y_3 = X_3 + \mathcal{P}(\text{Shuffle}(M_{\text{RFG}})). \quad (24)$$

This formulation connects the bottleneck frequency descriptor to multi-scale spatial refinement at $H/16$ resolution.

Progressive Decoder and Skip Fusion

After frequency-guided Stage-3 refinement, the decoder progressively restores spatial resolution using bilinear upsampling and depthwise-separable convolution.

The first refined decoder feature has 384 channels, $D_4 = Y_3$.

The subsequent decoder stages follow Eqs. (5)–(7).

For a 352×352 input, the decoder dimensions are $D_4 : 384 \times 22 \times 22$, $D_3 : 192 \times 44 \times 44$, $D_2 : 96 \times 88 \times 88$, and $D_1 : 96 \times 176 \times 176$.

At the F_2 and F_1 skip levels, the upsampled decoder representation is concatenated with the corresponding encoder feature and processed by a lightweight fusion block containing a 1×1 projection followed by a depthwise-separable 3×3 convolution.

No additional attention mechanism, frequency module, MSCB, or boundary-specific refinement block is applied at these decoder stages.

The final decoder stage has no direct encoder skip connection because the ConvNeXt stem begins at one-quarter of the input resolution.

Prediction Head

The final decoder feature D_1 is processed using a lightweight refinement head. Two depthwise-separable 3×3 convolutions transform the feature dimensions as $96 \rightarrow 96 \rightarrow 48$.

Each transformation is followed by Layer Normalization and GELU activation. A final 1×1 convolution produces a single-channel segmentation logit map:

$$P_0 = \text{Conv}_{1 \times 1}(\mathcal{R}(D_1)), \quad (25)$$

$$P = \text{Up}_{H,W}(P_0), \quad (26)$$

$$\hat{Y} = \sigma(P). \quad (27)$$

The prediction in Eq. (26) is bilinearly resized from $H/2 \times W/2$ to the original input resolution. The network directly returns logits; during inference, Eq. (27) gives the foreground probability map,

where $\sigma(\cdot)$ denotes the sigmoid function.

The selected architecture does not employ auxiliary prediction heads or deep supervision.

Model Complexity

The complete model has 30.06 M trainable and total parameters. Profiling with THOP on a $1 \times 3 \times 352 \times 352$ input gives 13.91 GMACs, corresponding to 27.82 GFLOPs under the project convention $\text{FLOPs} = 2 \times \text{MACs}$. Table 1 places these measurements alongside complexity values that could be verified from original publications or official implementations.

Among rows with verified parameter counts, the model is smaller than the Res2Net-50 variant of MEGANet. The available evidence is too sparse to rank all methods by parameter count or computational cost: several publications do not expose both values in accessible primary material, and DVFIP-Net’s reported GFLOPs lack a verified input resolution and counting convention. The final model therefore has a moderate parameter count within the verifiable subset, but its 27.82-GFLOP value should not be interpreted as directly higher or lower than competitor values measured under unspecified protocols.

The additional frequency-guided mechanism is introduced only at the first decoder–encoder fusion stage. The remaining decoder retains depthwise-separable convolution and conventional skip fusion, limiting the architectural modification to the stage at which high-level semantic information is first combined with the relatively high-resolution F_3 representation.

Table 1: Computational complexity comparison. Parameter counts and computational cost are taken from original publications or official implementations where available. Reported GFLOPs may use different input resolutions and counting conventions; ‘–’ denotes an unavailable or unverified value.

Method	Year	Input	Params (M)	Reported GFLOPs
CTNet [8]	2024	352 ²	–	–
MEGANet (Res2Net-50) [17]	2024	352 ²	44.19	–
Polyp-Mamba [16]	2025	–	–	–
MEIN [19]	2025	–	–	–
MF-Net [11]	2025	–	–	–
MSBP-Net [18]	2026	–	–	–
CIFFormer [9]	2025	–	–	–
PFPRNet [10]	2025	–	–	–
PraNet-V2 (Res2Net-50) [20]	2025	352 ²	–	–
MLB-Net [21]	2026	–	–	–
MSPN [22]	2026	–	–	–
DVFIP-Net [23]	2026	–	–	19.37
Ours	–	352 ²	30.06	27.82[†]

[†]Our value uses 13.91 GMACs and the explicit convention FLOPs = $2 \times$ MACs. Competitor values are preserved in source terminology and are not convention-normalized.

Experiments

Datasets and Data Splits

To evaluate the proposed framework under a fixed training and evaluation design, we used Kvasir-SEG [24], CVC-ClinicDB [25], CVC-300 [26], CVC-ColonDB [27], and ETIS-LaribPolypDB [28]. Kvasir-SEG and CVC-ClinicDB were split independently with a 90:10 ratio using shuffled sampling with random seed 42. Their training subsets were merged, giving 900 Kvasir-SEG and 550 CVC-ClinicDB images for training. The corresponding held-out subsets of 100 and 62 images were merged for validation and were also evaluated separately for reporting. CVC-300, CVC-ColonDB, and ETIS-LaribPolypDB were used only for external evaluation; none influenced training or checkpoint selection.

Table 2: Dataset use in the final experiment.

Dataset	Total	Train	Validation	External test
Kvasir-SEG	1000	900	100	–
CVC-ClinicDB	612	550	62	–
CVC-300	60	–	–	60
CVC-ColonDB	380	–	–	380
ETIS-LaribPolypDB	196	–	–	196

Images were converted to RGB, resized to 352×352 with bilinear interpolation, and scaled to $[0, 1]$. Masks were resized with nearest-neighbor interpolation and binarized. The encoder applied ImageNet normalization with mean (0.485, 0.456, 0.406) and standard deviation (0.229, 0.224, 0.225).

Implementation Details

The three final runs used seeds 42, 6543, and 7777. All final and ablation results are reported as the mean $\pm$ sample standard deviation (ddof=1) across these seeds unless otherwise noted. The runs were implemented in PyTorch 2.5.1+cu124 with a maximum of 200 epochs and batch size 24. AdamW with betas (0.9, 0.999) used separate parameter groups. The maximum learning rates were 3.0×10^{-4} for the decoder and 4.5×10^{-4} for the final refinement head. Encoder Stages 4, 3, 2, and 1 plus the stem used 3.0×10^{-5} , 2.4×10^{-5} , 1.92×10^{-5} ,

and 1.536×10^{-5} , respectively, corresponding to layer-wise decay 0.8. Weight decay was 1.0×10^{-4} for the decoder, 5.0×10^{-2} for the encoder, and 1.0×10^{-2} for the refinement head.

Learning rates were warmed up linearly for five epochs from 0.1 to 1.0 of each group maximum and then cosine-annealed over the remaining epochs to 1.0×10^{-6} . Training used AMP FP16, global gradient clipping at norm 1.0, and an exponential moving average (EMA) with decay 0.995. All parameters were trainable from the first epoch; no staged freezing or decoder-only warmup was used. The encoder was initialized from `convnext_tiny_22k_1k_384.pth`, logged as ImageNet pretrained weights. Deep supervision and frequency augmentation were disabled, and sampling was uniform. Early stopping was configured with patience 30 from the start of training. Seeds 42 and 6543 completed the maximum 200 epochs; seed 7777 stopped early at final epoch 167. Model complexity and its measurement convention are reported in Table 1.

Training augmentation applied horizontal and vertical flips independently with probability 0.5, rotation in $[-15^\circ, 15^\circ]$ with probability 0.5, and a random crop ratio in $[0.7, 1.0]$ followed by resizing with probability 0.5. Joint brightness and contrast factors in $[0.85, 1.15]$ and saturation in $[0.9, 1.1]$ were applied with probability 0.5. A 3×3 Gaussian blur and additive Gaussian noise with standard deviation 0.015 were applied with probabilities 0.15 and 0.25, respectively.

Loss Function

The final objective was

$$\mathcal{L} = 0.55\mathcal{L}_{\text{Dice}} + 0.25\mathcal{L}_{\text{BCE}} + 0.20\mathcal{L}_{\text{boundary}}. \quad (28)$$

The objective in Eq. (28) uses a soft Dice term with a smoothing constant of 1.0. The BCE-with-logits term used targets smoothed as $y_{\text{BCE}} = 0.98y + 0.01$, corresponding to label smoothing 0.02. The boundary-aware term used $\kappa = 5$. The available focal-Tversky term had weight zero and therefore did not contribute to training.

Evaluation Metrics

We report mean Dice (mDice), mean intersection over union (mIoU), S-measure S_α [29], weighted F-measure F_β^w , mean and maximum E-measure (mE_ϕ and $\max E_\phi$) [30], and mean absolute error (MAE) [31]. For each image, the thresholded prediction P and binary ground truth G were flattened and evaluated as

$$\text{Dice}(P, G) = \frac{2|P \cap G| + 10^{-6}}{|P| + |G| + 10^{-6}}, \quad \text{IoU}(P, G) = \frac{|P \cap G| + 10^{-6}}{|P| + |G| - |P \cap G| + 10^{-6}}. \quad (29)$$

The reported mDice and mIoU are arithmetic means of the per-image scores in Eq. (29), rather than global pixel aggregates. S-measure, weighted F-measure, the E-measure curve, and MAE were computed per image with `py_sod_metrics` from 8-bit probability maps and binary ground truth. The mean and maximum of the resulting E-measure curve were reported.

Model Selection and Evaluation Protocol

At each epoch, EMA weights were evaluated on the merged held-out Kvasir-SEG and CVC-ClinicDB validation set. Probabilities were obtained with one sigmoid operation and binarized using the strict rule $p > 0.45$. A checkpoint was saved only when validation IoU was strictly greater than the previous best. The selected checkpoints were epochs 184, 175, and 137 for seeds 42, 6543, and 7777, respectively. Final evaluation loaded each best checkpoint strictly and evaluated the two held-out subsets separately, followed by the full CVC-300, CVC-ColonDB, and ETIS-LaribPolypDB sets. Test-time augmentation was disabled, and external test results did not affect checkpoint selection.

Experimental Study

The experimental study first examines the contribution of the architectural components under controlled variants and then compares the resulting model with published methods. Presenting the ablation before the comparative evaluation establishes the reference configuration and clarifies which part of the final design is being assessed in the comparative tables.

Ablation Study

We first examine the contribution of the bottleneck enhancement, multi-scale refinement, and residual frequency guidance under the common experimental protocol. The variants differ only in their architectural components, allowing the final model to be interpreted relative to a sequence of increasingly expressive reference configurations.

Ablation Results

The sequence begins with FAFEM as the bottleneck enhancement module. The FAFEM-only configuration contains no multi-scale refinement layers and provides the final-protocol reference. A no-FAFEM baseline under the final training protocol is not available; older baseline measurements used a different protocol and are therefore excluded from this comparison.

Adding a Stage-3 MSCB-lite layer introduces parallel multi-scale refinement without frequency guidance. This configuration supplies the appropriate reference for the proposed guided variant because it preserves the multi-scale path while omitting the cross-level conditioning. The final configuration then adds residual frequency-guided coupling to that Stage-3 layer. The complete numerical comparison is given in Table 3; these results are descriptive and no significance test was performed.

A single-seed direct frequency-guidance configuration tested the mechanism without residual coupling. Its Kvasir-SEG result did not exceed the Stage-3 MSCB-lite result in the paired seed-42 comparison, so this experiment does not isolate the independent effects of residualization and frequency conditioning.

Table 3: Ablation study on Kvasir-SEG.

Method	Params (M)	mDice	mIoU	MAE
FAFEM	29.40	0.926 ± 0.002	0.876 ± 0.003	0.025 ± 0.0004
FAFEM + MSCB	30.00	0.930 ± 0.002	0.881 ± 0.003	0.024 ± 0.0004
Ours	30.06	0.931 ± 0.004	0.883 ± 0.006	0.024 ± 0.001

Ablation Interpretation

The ablation supports a staged interpretation of the design. FAFEM establishes the feature-enhanced reference, MSCB-lite adds multi-scale refinement at Stage 3, and RFG-MSCB introduces adaptive branch modulation on top of that reference path. The increment associated with the final coupling is small, so the ablation supports the feasibility of the proposed refinement but not a claim of broad or statistically superior improvement. The parameter increase shown in the table is correspondingly the cost of adding the refinement components rather than evidence that the full architecture is uniformly preferable.

Stage Placement and Generalization

The placement of the multi-scale refinement layer was also examined experimentally. In the final configuration, it is applied only at encoder Stage 3, a choice selected from the available ablation evidence rather than from a theoretical assumption. A single-seed exploratory all-skip configuration used batch size 32 and the same learning rate; the available training evidence did not improve relative to the Stage-3 configuration. Because the run was incomplete and differed in batch size, it does not support a general conclusion about optimal placement.

On the four additional datasets, the change from FAFEM + MSCB to the final guidance configuration was mixed in the paired seed-42 evidence: mDice decreased on CVC-ClinicDB and CVC-300 and increased on CVC-ColonDB and ETIS. These comparisons do not establish uniform cross-dataset improvement. The residual-coupling configuration with default initialization did not complete training and is therefore excluded from the quantitative ablation; initialization sensitivity remains a subject for future work.

Comparative study

After the component-level analysis, the comparative study evaluates the complete model on the held-out and external datasets and places its performance alongside published baselines. The quantitative results are presented first, followed by the qualitative analysis section reserved for visual evidence.

Quantitative analysis

Using the reporting convention defined in Implementation Details, the method obtained mDice/mIoU of $0.931 \pm 0.004 / 0.883 \pm 0.006$ on Kvasir-SEG and $0.937 \pm 0.002 / 0.888 \pm 0.004$ on CVC-ClinicDB. In the displayed comparisons, its Kvasir-SEG mDice was 0.17 percentage points below CAFÉ-Net, while its mIoU was 0.64 percentage points below CAFÉ-Net and higher than the remaining listed methods. On CVC-ClinicDB, PFPRNet exceeded the model by 1.20 percentage points in mDice and 1.53 percentage points in mIoU.

Across the comparison tables, the best and second-best values are shown in bold and underlined, respectively.

On the external datasets, mDice/mIoU were $0.882 \pm 0.002 / 0.810 \pm 0.002$ on CVC-300, $0.794 \pm 0.009 / 0.714 \pm 0.009$ on CVC-ColonDB, and $0.787 \pm 0.011 / 0.707 \pm 0.012$ on ETIS-LaribPolypDB. The model did not attain the highest displayed external-dataset mean. Its largest between-seed variation in mDice and mIoU occurred on ETIS. These comparisons are descriptive; no statistical significance test between methods was performed.

Table 4: Quantitative comparison on Kvasir-SEG.

Methods	Year	mDice	mIoU	F_{β}^w	S_{α}	mE_{ξ}	$\max E_{\xi}$	MAE
CTNet [8]	2024	0.917	0.863	0.910	0.928	0.959	0.964	0.023
MEGANet [17]	2024	0.913	0.863	0.907	0.918	0.956	0.959	0.025
CAFÉ-Net [32]	2024	0.933	0.889	0.927	<u>0.939</u>	0.967	–	0.019
Polyp-LVT [33]	2024	0.909	0.851	0.913	0.922	0.941	–	0.024
Polyp-Mamba [16]	2025	0.919	0.867	0.912	0.951	–	<u>0.968</u>	<u>0.021</u>
MEIN [19]	2025	0.905	0.847	0.905	0.914	–	0.964	0.026
MF-Net [11]	2025	0.915	0.863	0.939	0.924	0.962	–	0.022
MSBP-Net [18]	2025	0.919	0.868	0.916	0.926	0.961	–	0.023
CIFFormer [9]	2025	0.926	0.876	<u>0.929</u>	<u>0.939</u>	0.967	–	0.019
PFPRNet [10]	2025	0.930	0.881	0.920	0.935	0.971	–	0.019
PraNet-V2 [20]	2025	0.907	0.853	–	–	–	–	0.025
MLB-Net [21]	2026	0.926	0.878	0.921	0.938	<u>0.969</u>	0.975	<u>0.021</u>
MSPN [22]	2026	0.927	0.879	–	–	–	–	<u>0.021</u>
DVFIP-Net [23]	2026	0.922	0.874	–	–	–	–	–
Ours	–	<u>0.931</u> <u>± 0.004</u>	<u>0.883</u> <u>± 0.006</u>	0.911 ± 0.005	0.933 ± 0.003	0.960 ± 0.003	0.967 ± 0.003	0.024 ± 0.001

Table 5: Quantitative comparison on CVC-ClinicDB.

Methods	Year	mDice	mIoU	F_{β}^w	S_{α}	mE_{ξ}	$\max E_{\xi}$	MAE
CTNet [8]	2024	0.936	0.887	0.934	0.952	<u>0.983</u>	0.986	0.006
MEGANet [17]	2024	0.938	0.894	0.940	0.950	–	<u>0.987</u>	<u>0.007</u>
Polyp-Mamba [16]	2025	0.941	0.896	0.936	0.970	–	0.970	0.008
MEIN [19]	2025	0.923	0.878	0.928	0.944	–	0.970	0.008
MF-Net [11]	2025	0.931	0.884	0.953	0.949	<u>0.983</u>	–	0.006
MSBP-Net [18]	2025	0.940	0.892	0.927	0.939	0.976	–	<u>0.007</u>
CIFFormer [9]	2025	0.944	0.897	0.941	0.934	0.972	–	<u>0.007</u>
PFPRNet [10]	2025	0.949	0.903	0.947	0.957	0.989	–	0.006
PraNet-V2 [20]	2025	0.929	0.880	–	–	–	–	0.009

MLB-Net [21]	2026	<u>0.945</u>	<u>0.902</u>	<u>0.948</u>	<u>0.958</u>	0.989	0.992	0.006
MSPN [22]	2026	0.943	<u>0.902</u>	–	–	–	–	0.006
DVFIP-Net [23]	2026	0.942	0.896	–	–	–	–	–
Ours	–	<u>0.937</u> ± 0.002	<u>0.888</u> ± 0.004	<u>0.925</u> ± 0.003	<u>0.947</u> ± 0.001	<u>0.978</u> ± 0.001	<u>0.983</u> ± 0.001	<u>0.011</u> ± 0.0003

Table 6: Quantitative comparison on ETIS-LaribPolypDB.

Methods	Year	mDice	mIOU	F_{β}^w	S_{α}	mE_{ξ}	$\max E_{\xi}$	MAE
CTNet [8]	2024	0.810	0.734	0.776	0.886	0.913	<u>0.921</u>	0.014
MEGANet [17]	2024	0.789	0.709	0.753	0.866	0.855	0.915	0.015
Polyp-Mamba [16]	2025	0.756	0.668	0.756	0.887	–	0.872	0.014
MEIN [19]	2025	0.788	0.711	0.764	0.870	–	0.902	0.018
MF-Net [11]	2025	<u>0.821</u>	<u>0.750</u>	<u>0.850</u>	<u>0.901</u>	<u>0.925</u>	–	0.011
MSBP-Net [18]	2025	0.795	0.718	0.770	0.873	0.899	–	0.019
CIFFormer [9]	2025	0.810	0.746	0.858	0.889	0.918	–	0.013
PFPRNet [10]	2025	0.828	0.764	0.839	0.908	0.935	–	<u>0.012</u>
PraNet-V2 [20]	2025	0.713	0.633	–	–	–	–	0.021
MLB-Net [21]	2026	<u>0.821</u>	0.744	0.780	0.899	0.914	0.935	0.013
MSPN [22]	2026	0.798	0.717	–	–	–	–	0.017
DVFIP-Net [23]	2026	0.811	0.730	–	–	–	–	–
Ours	–	<u>0.787</u> ± 0.011	<u>0.707</u> ± 0.012	<u>0.753</u> ± 0.007	<u>0.878</u> ± 0.004	<u>0.902</u> ± 0.002	<u>0.916</u> ± 0.003	<u>0.015</u> ± 0.001

Table 7: Quantitative comparison on CVC-ColonDB.

Methods	Year	mDice	mIOU	F_{β}^w	S_{α}	mE_{ξ}	$\max E_{\xi}$	MAE
CTNet [8]	2024	0.813	0.734	0.801	0.874	0.915	<u>0.919</u>	0.040
MEGANet [17]	2024	0.793	0.714	0.779	0.854	0.892	0.895	0.034
Polyp-Mamba [16]	2025	0.791	0.713	0.765	0.846	–	0.902	0.034
MEIN [19]	2025	0.791	0.715	0.785	0.856	–	0.897	0.034
MF-Net [11]	2025	0.825	<u>0.751</u>	0.864	0.887	0.929	–	<u>0.024</u>
MSBP-Net [18]	2025	0.810	0.731	0.810	0.861	0.912	–	0.029
CIFFormer [9]	2025	0.823	0.741	<u>0.851</u>	0.874	0.918	–	0.026
PFPRNet [10]	2025	0.819	0.741	0.843	0.882	0.915	–	0.028
PraNet-V2 [20]	2025	0.702	0.620	–	–	–	–	0.044
MLB-Net [21]	2026	0.832	0.757	0.825	<u>0.886</u>	<u>0.924</u>	0.931	0.023
MSPN [22]	2026	0.811	0.734	–	–	–	–	0.029
DVFIP-Net [23]	2026	<u>0.826</u>	0.746	–	–	–	–	–
Ours	–	<u>0.794</u> ± 0.009	<u>0.714</u> ± 0.009	<u>0.772</u> ± 0.007	<u>0.861</u> ± 0.006	<u>0.898</u> ± 0.004	<u>0.905</u> ± 0.004	<u>0.035</u> ± 0.001

Table 8: Quantitative comparison on CVC-300 (EndoScene).

Methods	Year	mDice	mIOU	F_{β}^w	S_{α}	mE_{ξ}	$\max E_{\xi}$	MAE
CTNet [8]	2024	<u>0.908</u>	0.844	0.894	0.975	0.975	0.982	<u>0.006</u>
MEGANet [17]	2024	0.899	0.834	0.882	0.935	0.966	0.969	0.007
Polyp-Mamba [16]	2025	0.906	0.840	0.888	0.936	–	0.975	<u>0.006</u>

MEIN [19]	2025	0.893	0.828	0.880	0.930	–	0.965	0.007
MF-Net [11]	2025	0.897	0.836	<u>0.918</u>	0.941	0.966	–	0.007
MSBP-Net [18]	2025	0.903	0.859	0.927	0.942	0.967	–	0.019
CIFFormer [9]	2025	0.883	0.822	0.912	0.939	0.961	–	0.100
PFPRNet [10]	2025	0.891	0.833	0.917	0.928	0.950	–	0.013
PraNet-V2 [20]	2025	0.873	0.795	–	–	–	–	0.007
MLB-Net [21]	2026	0.915	<u>0.852</u>	0.909	<u>0.974</u>	<u>0.974</u>	<u>0.978</u>	0.005
MSPN [22]	2026	0.907	0.840	–	–	–	–	<u>0.006</u>
DVFIP-Net [23]	2026	0.900	0.833	–	–	–	–	–
Ours	–	0.882 ± 0.002	0.810 ± 0.002	0.857 ± 0.001	0.928 ± 0.003	0.956 ± 0.005	0.971 ± 0.002	0.007 ± 0.0003

Discussion

Quantitative and Ablation Interpretation

The comparative results indicate strong performance on the two held-out datasets, whereas performance is less consistent on the external datasets. The complete numerical results and uncertainty estimates are reported in the quantitative-analysis subsection and comparison tables. This pattern is consistent with the architectural motivation: the model retains a convolutional multi-scale pathway for efficient decoding while adding sample-dependent guidance intended to adapt that pathway to image content.

The ablation provides a more specific interpretation of the contribution. FAFEM establishes the bottleneck feature-enhancement reference, MSCB-lite adds multi-scale refinement at Stage 3 without frequency guidance, and RFG-MSCB adds adaptive branch modulation to that reference path. The final increment is small, indicating that the proposed coupling is a controlled refinement rather than a wholesale change in the decoder. The paired single-seed comparison with direct frequency guidance also did not show an advantage without residual coupling; consequently, the available evidence does not disentangle residualization from frequency conditioning. These observations support the feasibility of the combined mechanism but leave its individual causal components unresolved.

The bounded residual formulation may explain why the added mechanism can be introduced without discarding the ordinary multi-scale response. By interpolating between uniform and predicted branch weights, it provides a conservative path from the established MSCB behavior to sample-dependent refinement. This is a post hoc interpretation rather than a separately tested mechanism-level conclusion, because the ablation measures the combined design.

Cross-Dataset Generalization

The model performs more consistently on the held-out datasets than on the external datasets, where differences in endoscopic systems and polyp-morphology distributions create a stronger domain shift from the training data. The guidance change is also not uniformly positive across the external datasets in the paired single-seed evidence. Thus, the results support transfer across the evaluated datasets only in a limited, descriptive sense and do not establish broad out-of-distribution generalization.

The weaker behavior on CVC-ColonDB and ETIS-LaribPolypDB, together with the observed variability on ETIS, suggests that domain adaptation or targeted augmentation may be needed. The model also does not outperform several recent methods on these datasets, which reinforces that the proposed frequency-guided refinement does not remove the underlying domain-shift challenge.

Practical Trade-offs and Placement

As summarized in Table 1, the final architecture has a moderate parameter count within the verifiable subset. Computational-cost rankings remain uncertain because most competitor reports do not provide a verified input resolution together with an explicit FLOP-counting convention. No matched-hardware latency benchmark was performed, so the present evidence does not establish real-time inference suitability.

The Stage-3 placement was selected from the available ablation evidence and keeps the new guidance localized to the first decoder–encoder fusion. The exploratory all-skip comparison was single-seed, incomplete, and used a different batch size, so it does not support a general conclusion about optimal placement. Together, the complexity

and placement results describe a constrained architectural trade-off, but they do not establish the best deployment configuration.

These interpretive limits motivate the following limitations and future-work priorities.

Limitations

- **No statistical significance testing:** All reported differences are descriptive; no hypothesis testing was performed.
- **Single-seed placement evidence:** The exploratory all-skip placement run is single-seed, incomplete, and uses a different batch size; it cannot establish a general placement rule.
- **No disentanglement of RFG-MSCB components:** The ablation isolates the combined residual frequency-guided coupling; individual contributions of the residual formulation, initialization strength, and frequency conditioning are not separately measured.
- **Incomplete experiments:** The residual-guidance run with default initialization did not complete training; its potential findings are unavailable.
- **Domain gap on external datasets:** Performance drops substantially on CVC-ColonDB and ETIS-LaribPolypDB, indicating limited out-of-distribution generalization.
- **Hardware and software not fully verified:** GPU model, CUDA/cuDNN versions, and `py_sod_metrics` version are not formally recorded.
- **Training duration:** Seed 7777 stopped early at epoch 167 (patience 30), while seeds 42 and 6543 reached epoch 199; this asymmetry may affect the mean estimate.

Future Work

- Systematic multi-seed placement ablation with controlled batch size and learning rate.
- Disentanglement of the RFG-MSCB components (residual form, initialization schedule, frequency conditioning) via factorial experiments.
- Investigation of initialization sensitivity using completed default-initialization runs.
- Domain adaptation or targeted augmentation to improve generalization on CVC-ColonDB and ETIS-LaribPolypDB.
- Statistical significance testing with larger seed ensembles or bootstrap confidence intervals.
- Formal recording of hardware, software, and dependency versions for full reproducibility.

Conclusion

Taken together, the experiments support a cautious conclusion. We presented a ConvNeXt-Tiny U-shaped segmentation framework that couples prior FAFEM and MSCB modules through RFG-MSCB at the Stage-3 encoder–decoder fusion. The design extends bottleneck frequency enhancement into decoder refinement while preserving the ordinary multi-scale pathway as a reference, thereby addressing the motivation for sample-dependent receptive-field selection without replacing the established decoder path.

The empirical evidence supports a cautious interpretation. The ablation indicates that the added residual frequency guidance produces a small change relative to the FAFEM + MSCB reference, while the external-dataset results are less consistent than the held-out results. The complexity comparison and localized placement describe a moderate architectural trade-off, but the available experiments do not establish uniform gains, component-level causality, or real-time suitability.

These findings motivate systematic multi-seed placement experiments, disentanglement of residualization and frequency conditioning, and domain-adaptation strategies for the more challenging external datasets. The incomplete default-initialization run, single-seed placement evidence, and absence of significance testing should be resolved before making stronger claims about generalization or superiority.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The datasets used in this study are publicly available from the sources cited in the manuscript. The project code, trained weights, and derived evaluation artifacts are maintained locally and are not currently available from a public repository.

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